



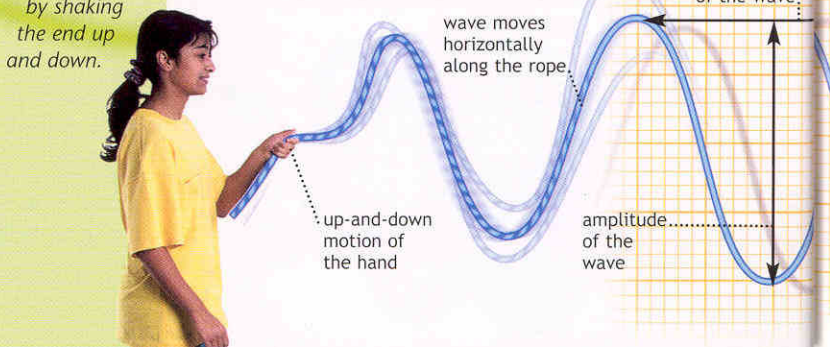
The Scottish physicist **James Clerk Maxwell** (1831–79) discovered the set of four equations that describe the behavior

of electromagnetic radiation in classical terms. This is the greatest achievement of classical physics besides the work of Isaac Newton. As well as developing the mathematical description of light, Maxwell came up with the “three-color” process of making colored images, which is used in domestic TV receivers today.

Maxwell had been able to condense everything that could be said about electric and magnetic fields in a set of just four equations, which became known as Maxwell's Equations. They perform the same role in field theory (the theory of electromagnetic forces) as Newton's Laws do in mechanics (the theory of particle behavior). Among other things, the equations describe what happens when a moving electric wave, which might be caused by an

electron jiggling up and down in an atom, travels through space. As the wave ripples along, the electric field at each point it passes changes. But a changing electric field creates a magnetic field. So the electric wave creates a magnetic wave that moves in step with it. But the magnetic wave creates an electric wave. What really happens is that a jiggling electron produces not a simple electric wave, but a paired electric and magnetic wave—an electromagnetic wave. There is no such thing as a pure electric or a pure magnetic wave.

making waves
Electromagnetic waves are made when charged particles, such as electrons, jiggle around. This is analogous to the way a wave can be made to run along a rope by shaking the end up and down.



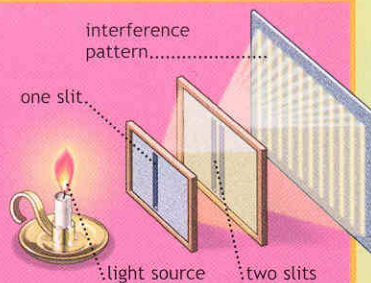
light as a wave

Two hundred years ago, Thomas Young demonstrated that light is a wave using an experiment with two holes. First, light with a pure color shines through a single, narrow slit



Thomas Young (1773–1829)

in a screen. The light from this hole passes through two parallel slits in a second screen, and falls on a third screen. There, it makes a pattern of light and dark stripes. The explanation is that light travels in a wave, like ripples in a pond. The ripples from each of the two slits go up and down, and where they meet they interfere. Where both ripples are in step, they make a brighter light; if they are exactly out of step, they cancel out and leave a dark stripe. It was an astonishing revelation to 19th-century scientists that adding two kinds of light together could make darkness.



making waves

Despite the simple structure of the equipment used by Young, the observed results were a revelation.

